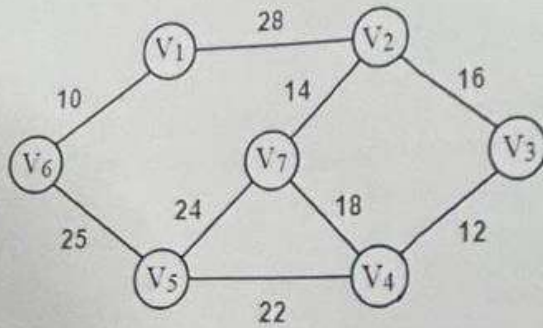




Problem # 2: Construct the Minimum Spanning Tree (MST) of the below graph using Kruskal's algorithm. Show steps. [6pts]





- 21) The total time in the system of the optimal schedule is:
- A. 94
 - B. 144
 - C. 150
 - D. None of the above

- 22) Based on the information given in the table below, the feasible sequence generated when solving the job scheduling with deadlines problem using the greedy method is:
- A. [1, 3, 4]
 - B. [2, 1, 4]
 - C. [1, 2, 3]
 - D. None of the above

Job	Deadline	Profit
1	2	100
2	1	70
3	2	60
4	3	30
5	2	10



Problem # 1: Multiple choice question. (22 x 1pc)

- 1) If $f(n) \in O(g(n))$ and $g(n) \in O(h(n))$, then $f(n) + g(n) \in$
- $O(g(n))$
 - $O(h(n))$
 - $O(\max\{g(n), h(n)\})$
 - None of the above

- 2) The best-case time complexity of sequential search is _____
- n
 - 1
 - n^2
 - $\log n$

- 3) Given the below sorted array:

$arr[] = \{4, 13, 17, 18, 19, 22, 30, 33, 34, 39, 46, 49, 80, 94, 110, 115, 117\}$

To determine if $x = 119$ exists in the above array using the binary search algorithm, the numbers compared in order with x are:

- 34, 80, 110, 115, 117
- 39, 49, 94, 115, 117
- 34, 18, 13, 17, 4
- None of the above

- 4) Minimum number of moves required to solve a tower of Hanoi problem with 4 disks is _____
- 7
 - 3
 - 15
 - 16

- 5) The every-case time complexity of Prime's algorithm is _____
- $\theta(n^2 * \lg n)$
 - $\theta(\lg n)$
 - $\theta(n)$
 - None of the above

- 6) Problems that can be solved in polynomial time are called _____
- Intractable
 - Tractable
 - Nondeterministic
 - None of the above

- 7) Which of the following problems are in Nondeterministic Polynomial (NP)?
- Hamiltonian circuit existence
 - knapsack problem
 - Graph coloring
 - All of the above



- 8) Which of the following is a possible solution to the 4-queens problem using backtracking?
- A. $\langle 1, 2 \rangle, \langle 2, 4 \rangle, \langle 3, 1 \rangle, \langle 4, 3 \rangle$
 - B. $\langle 1, 1 \rangle, \langle 2, 1 \rangle, \langle 3, 2 \rangle, \langle 4, 1 \rangle$
 - C. $\langle 1, 3 \rangle, \langle 2, 1 \rangle, \langle 3, 4 \rangle, \langle 4, 2 \rangle$
 - D. (A + C)

- 9) The number of nodes in the state space tree to the m -coloring problem using backtracking is
- A. $m^n - 1$
 - B. $\frac{m^{n+1} - 1}{m - 1}$
 - C. $m - 1$
 - D. $n^m - 1$

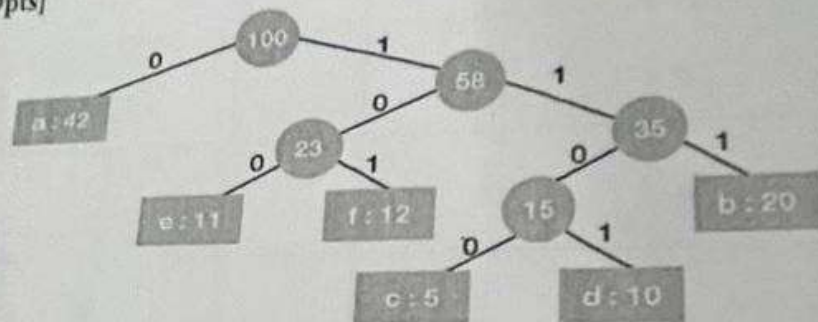
***** Answer the questions 10 to 12 for the given below recurrence equation:

$$t_n - t_{n-1} - 6t_{n-2} = 0$$

$$t_0 = 0, t_1 = 10$$

- 10) What is the characteristic equation?
- A. $r^2 - r + 6 = 0$
 - B. $r^2 + 6r - 1 = 0$
 - C. $r^2 - r + 6 = 0$
 - D. None of the above
- 11) What are the roots of the characteristic equation?
- A. $r = -3$ and $r = 2$
 - B. $r = 3$ and $r = -2$
 - C. $r = -3$ and $r = -2$
 - D. None of the above
- 12) What is the particular form of the solution of the recurrence equation?
- A. $t_n = 2 * (-3)^n - 2 * (2)^n$
 - B. $t_n = 2 * (3)^n - 2 * (-2)^n$
 - C. $t_n = 2 * (-3)^n - 2 * (-2)^n$
 - D. None of the above
- 13) Prime's algorithm is a
- A. Divide and conquer algorithm
 - B. Greedy algorithm
 - C. Dynamic programming algorithm
 - D. None of the above
- 14) The average-case time complexity of Quicksort search is
- A. $\theta(n * \lg n)$
 - B. $\theta(n)$
 - C. $\theta(n^2)$
 - D. None of the above

Problem # 5: Based on the below binary tree for Huffman code, Answer the following questions. [9pts]



A- Fill in the table below with the codeword for each character. [6 x 0.5pt]

Character	Code
a	
b	
c	
d	
e	
f	

B- What is the total number of bits used to store the file? [2pts]



Problem # 3: Given the two sequences "BDCABA" and "ABCBDAB", Complete the table below to find the Longest Common Sequence (LCS) using the Dynamic Programming (DP). [6pts]

j		0	1	2	3	4	5	6
i			B	D	C	A	B	A
	0	0	0	0	0	0	0	0
1	A	0	↑ 0	↑ 0	↑ 0	↖ 1	← 1	↖ 1
2	B	0	↖ 1	← 1	← 1	↑ 1	↖ 2	← 2
3	C	0						
4	B	0						
5	D	0						
6	A	0						
7	B	0						

LCS is

Problem # 4: Draw the entire pruned state space tree that is generated by the backtracking algorithm for the Sum of Subsets problem on the following instance: [6pts]

$$W = 13, w_1 = 3, w_2 = 4, w_3 = 5, w_4 = 6$$

C- Compared to the fixed-length binary code, how many bits can be saved when using Huffman code technique? [2pts]

D- What is the sequence of characters corresponding to the following code "111001001011101"? [2pts]

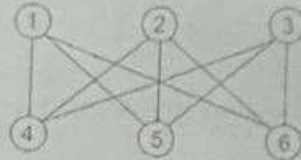


15) Which of the following is an algorithm to find the minimum spanning tree of the given graph?

- A. Prime's algorithm
- B. Kruskal's algorithm
- C. (A + B)
- D. None of the above

16) Which of the following is a possible Hamiltonian circuit of the below graph?

- A. {1, 6, 3, 4, 1}
- B. {1, 4, 2, 6, 3, 5, 1}
- C. {1, 4, 3, 6, 2, 4, 1}
- D. None of the above



17) Which of the following is the correct order of growth (from the lowest to highest)?

- A. $\theta(\lg n)$, $\theta(n)$, $\theta(n \lg n)$, $\theta(n^5)$, $\theta(n^6)$, $\theta(4^n)$, $\theta(5^n)$, $\theta(n^n)$, $\theta(n!)$
- B. $\theta(\lg n)$, $\theta(n)$, $\theta(n \lg n)$, $\theta(n^5)$, $\theta(n^6)$, $\theta(4^n)$, $\theta(5^n)$, $\theta(n!)$, $\theta(n^n)$
- C. $\theta(\lg n)$, $\theta(n)$, $\theta(n \lg n)$, $\theta(n^5)$, $\theta(n^6)$, $\theta(5^n)$, $\theta(4^n)$, $\theta(n!)$, $\theta(n^n)$
- D. None of the above

18) Problems on which Greedy algorithm approach work has a property:

- A. Greedy-choice property
- B. Optimal substructure
- C. (A + B)
- D. None of the above

19) Which of the following is a basic component in Greedy approach?

- A. Selection procedure
- B. Feasibility check
- C. Solution check
- D. All of the above

**** Suppose there are five jobs and the service time of these jobs are $t_1 = 10$, $t_2 = 15$, $t_3 = 5$, $t_4 = 8$, and $t_5 = 3$, answer the questions 20 and 21:

20) The schedule that minimizes the total time in the system is:

- A. [1, 2, 3, 4, 5]
- B. [2, 1, 4, 3, 5]
- C. [5, 3, 4, 1, 2]
- D. None of the above